Probing, scattering, and dimensional extensions to the Systrophē framework

Five further modules (Phases 21–25)

Christian Knopp
cknopp@gmail.com

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Abstract

We add five further modules to the Systrophē framework covering probe diagnostics, wave scattering, numerical-relativity hand-off, dark-matter coupling, and higher-dimensional embedding: (1) tidal-force scalar across CTC bands, including geodesic-deviation evolution and a safety-radius API; (2) Klein-Gordon scattering with transmission/reflection coefficients across chronology horizons, including a superradiance probe; (3) NR initial-data export (ADM + BSSN) with constraint-residual diagnostics and 3D Cartesian extrusion; (4) dark-matter scalar coupling treating the cylinder as an ultra-light DM soliton anchor; (5) 5D Kaluza-Klein embedding showing that motion in a compact extra dimension can convert a 4D CTC into a 5D spacelike path. Seventy-one new tests across the five modules, all passing. Code at `github.com/Zynerji/systrophe` (v0.19.0+).

1 Tidal forces across CTC bands (Phase 21)

A probe near a chronology horizon experiences strong tidal forces. The dominant radial tidal scalar is

$$T(r) = -\tfrac{1}{2}\, F''(r),$$

which oscillates log-periodically and diverges as r approaches each F = 0 surface.

The module `tidal_forces.py` provides:

- `riemann_scalar_radial`: $T(r)$;
- `tidal_stretch_at_radius`, `tidal_squeeze_at_radius`;
- `geodesic_deviation_evolution`: Jacobi equation integration;
- `multi_band_tidal_signature`: T at the midpoint of each CTC band;
- `safety_radius_for_probe`: where $|T|$ exceeds a material strength.

For omega = R = 1, $|T|$ stays small in the first CTC band but grows log-periodically toward the second, then larger horizons – a probe made of any finite material has a definite stay-out radius.

2 Klein-Gordon scattering (Phase 22)

A massive scalar ϕ on the LP exterior obeys $(\Box - m^2)\phi = 0$. Decomposing $\phi = e^{-i\omega t} e^{in\phi}\chi(r)$, the radial equation reduces to a Schrödinger-like form with effective potential

$$V_{\text{eff}}(r; \omega, n, m) = \frac{\omega^2}{F_{\text{eff}}(r)} - m^2 - \frac{n^2}{|L|}.$$

The module `kg_scattering.py` provides:

- `effective_potential`;
- `transmission_coefficient`: WKB $|T|^2 = \exp\Big(-2\int \sqrt{|V_{\text{eff}}|}\, dr\Big)$;
- `reflection_coefficient`: $|R|^2 = 1 - |T|^2$;
- `transmission_at_multiple_CH`: cumulative transmission;
- `superradiance_test`: Kerr-analog amplification condition $\omega < n\,\Omega_H$;
- `bound_states_between_CHs`: discrete spectrum in CTC bands.

Each chronology horizon acts as a partial barrier; cumulative transmission decreases geometrically across multiple CHs. Modes with $\omega < n\,\Omega_H$ are amplified – the LP analog of Kerr superradiance.

3 NR initial-data export (Phase 23)

The module `nr_initial_data.py` provides constraint-checked hand-off to numerical-relativity codes (Einstein Toolkit, GRChombo, BAM):

- `adm_profile_1d`: lapse, shift, spatial metric, extrinsic curvature on a 1D radial profile;
- `hamiltonian_constraint_residual` and `momentum_constraint_residual`: vacuum-constraint residuals;
- `bssn_decomposition`: conformal factor χ, conformal metric $\tilde{\gamma}$, traceless $\tilde{A}$;
- `extrude_to_3d_cartesian`: 3D grid for NR import;
- `export_initial_data_json`: JSON serialization;
- `cauchy_horizon_warning_for_foliation`: detect lapse breakdown.

This makes the Systrophē framework directly importable into production NR codes. The Cauchy-horizon warning ensures that the foliation choice is safe before evolution starts.

4 Dark-matter scalar coupling (Phase 24)

For an ultra-light dark-matter scalar ($m_{\text{DM}} \sim 10^{-22}$ eV fuzzy-DM regime), the LP cylinder anchors a soliton-like coherent field. We model:

$$\rho_{\text{DM}}(r) = \frac{\rho_c}{(1 + (r/r_{\text{sol}})^2)^8}$$

with r_{sol} set by the Compton wavelength of m_{DM} and the cylinder mass.

The module `dm_scalar_coupling.py` provides:

- `compton_wavelength`;
- `DM_density_profile_around_cylinder`;
- `DM_drag_on_orbits`: orbital-frequency shift;
- `DM_superradiance_growth_rate`: Kerr-analog;
- `effective_alpha_with_DM_pressure`;
- `DM_induced_CTC_lifetime`: DM-noise decoherence time.

For realistic ULDM densities, the DM-noise CTC lifetime exceeds the Hubble time – so even pessimistic DM scenarios do not erase existing Systrophē CTCs.

5 5D Kaluza-Klein embedding (Phase 25)

Embed the 4D LP metric in 5D with a compact ξ of radius L_ξ:

$$ds_5^2 = ds_{\mathrm{LP}}^2 + L_\xi^2 \, d\xi^2, \qquad \xi \sim \xi + 2\pi.$$

KK reduction yields a $U(1)$ gauge field plus a dilaton from $g_{\xi\xi}$. A worldline that moves rapidly in ξ acquires a positive 5D contribution that can offset a negative 4D one – *thereby converting a 4D CTC into a 5D spacelike path.*

The module `kk_embedding.py` provides:

- `five_d_metric`: 5×5 tensor;
- `kk_tower_masses`: $m_n = n/L_\xi$;
- `kk_reduced_alpha`: $\alpha\sqrt{1 + (R/L_\xi)^2}$;
- `xi_circulation_evades_CTC`: minimum $d\xi/d\phi$ to achieve a spacelike 5D path inside a 4D CTC band;
- `kk_monopole_charge`: $U(1)$ charge from non-trivial fiber;
- `compactification_radius_from_observations`: bound L_ξ;
- `five_d_geodesic_drift_in_xi`: per-revolution ξ drift.

Headline: in any 4D CTC band ($F < 0$), the minimum $d\xi/d\phi$ for a 5D-spacelike escape is $\sqrt{-F}/L_\xi$. Sub-Compton L_ξ requires impractical ξ velocities; macroscopic L_ξ would already have been observed in collider experiments. So KK escape is theoretically possible but phenomenologically gated.

6 Implications

The five modules extend the framework to:

1. **Probe physics**: tidal forces give a probe-survivability map.
2. **Wave scattering**: each chronology horizon is a quantum potential barrier with explicit transmission.
3. **NR pipeline**: initial-data is exportable to Einstein Toolkit / GRChombo with constraint residuals tagged.
4. **Cosmology / DM**: ULDM coupling is weak enough that existing CTCs survive Hubble-time DM noise.
5. **Higher dimensions**: KK extra-dim motion is a potential CTC escape route, gated by observational bounds on L_ξ.

7 Reproduction

Phase	Source file
21. Tidal forces	`tidal_forces.py`
22. KG scattering	`kg_scattering.py`
23. NR initial data	`nr_initial_data.py`
24. DM scalar coupling	`dm_scalar_coupling.py`
25. 5D KK embedding	`kk_embedding.py`

All 71 new tests pass; combined Systrophē suite at 813 passing tests.

References

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